

Another Russian company that creates exciting business solutions is Etersoft. It sells its boxed solutions built upon a heavily redefined WINE package. Thus, one of the most promising products is SELTA@Etersoft—a universal SQL-queries translator from T-SQL into pgSQL. Such transformation allows the use of the PostgreSQL server for a number of applications designed purely to work with Microsoft SQL Server only. So how does the translator work? It parses requests to MS SQL system tables, and adapts them transparently to PostgreSQL notation. So you don't need to interfere either with the server or the client program. Many companies that can't completely transfer their business onto a Linux platform (due to Microsoft SQL Server or Win32-applications) choose SELTA@Etersoft and WINE@Etersoft and happily launch Win32-applications within GNU/Linux environments.

Now let me describe the problems that occur with GNU/Linux once it is deployed in real life by Russian companies. The main problem is the quality and experience of the technical personnel, who're primary responsible for the transformation project—from Windows to Linux. The vast majority have never faced any UNIX system before. Therefore, they have great difficulty orienting themselves to the UNIX philosophy and they follow one simple principle—dumb replacement of one Windows GUI with a similar-looking GUI, but under the Linux logo. Naturally, the features that make Linux a truly unique system aren't reclaimed and aren't used at all.

The second reason is a difficult-to-eradicate myth about the importance of a nationwide distribution. If there had been one created earlier, with Russian (Hindi, Brazilian, you name it) localisation and other bells and whistles, it would be distributed in a blink of an eye. However, the long history of AltLinux shows another trend—there could be, de-facto, an already existing national distribution. In our case, AltLinux has no rudiments of the original MandrakeLinux because AltLinux was completely redesigned 10 years ago. There already exists a great channel to spread AltLinux both on CDs/DVDs, as well as multiple tutorials written by AltLinux authors for teaching and learning purposes, and there's great support from the company. But there's still Windows domination. Why? On the one hand, there is strong pressure from both the Microsoft side and from the financial department of any enterprise that decides to purchase Win32/Win64 products only. This can be explained only by this fact—real ignorance of whatever systems exist outside the Microsoft world. However, even when decision makers are aware of Linux/UNIX, their reason is usually this one: "We have specialists familiar with Microsoft, but we haven't any Linux specialists. Therefore, we take Microsoft." Strange, isn't it? Sadly, the truth is that sometimes, companies do not even have an IT department with sufficient Microsoft experience.

Another big problem is a strong existence of other myths (this is particularly evident with MCBC Linux): "There exists a purely native (Russian/Indian/German) Linux innovation and only selected specialists are keen to understand it." This problem

is really a huge one, because when some kind of problem arises, there's usually a solution outside of this closed world. For example, you should try to find fixes for Ubuntu packages not within the Ubuntu world only, but among Debian users too!

The third big problem is that engineering and IT personnel training is done with Microsoft products. This situation is rather typical for middle schools, as well as for higher education institutions. The courses devoted to UNIX topics are organised and created mostly by enthusiasts or organised by big vendors—so such training can be considered as supplementary to the selling channel of Linux software and middleware. The education policy in this aspect is still driven by the country's educational agency.

But the real surprise comes from a place no one expected before. First, Russia has a national search giant—Yandex. It uses Linux for its own purposes—server infrastructure. So it's allocated its resources to maintain the huge repository of core Linux distributions: AltLinux, ArchLinux, ASPLinux, CentOS, Debian, Ubuntu, Slackware, Mandriva, and others. It has also provided space for such office suites as OpenOffice, LibreOffice and Mozilla. They mirror kernel.org, VLC, MySQL, KDE, GNOME and some others. From this perspective, the national software repository is already here. Moreover, the agreements on free peering are achieved with a number of local ISPs throughout the whole country. This fact alone creates an ideal opportunity for sharing any distro. In addition to free access to the Yandex repository, it also provides free hosting facilities to video, pictures, Web hosting, etc. So there's no problem in distributing any books, presentations or videos on Linux.

SMB-sized companies that have Linux-familiar IT staff silently migrate their network infrastructure onto Linux/FreeBSD/OpenSolaris, because its mission-critical nature allows firms to have 24x7 business. Occasionally, the government sector behaves similarly, unless there's particular Win32 software imposed from the federal government level. Having IT infrastructure based on Debian, Ubuntu or CentOS allows a government institution to save money, and instead of spending on Microsoft products, it can purchase more hardware or order custom software. The bigger advantages are from professionals that come to the government sector from commercial companies. They bring experience and knowledge with them and that is most important—the desire to change the infrastructure to be the one 'right' standard. Unfortunately, this is a temporary scenario, due to the financial crisis and other local factors. Once the business situation improves, such personnel return to the private sector where wages are usually higher.

In Russia, there exists an analogy to the Linux Foundation, known as the Russian Association for Free Software (RASFO). The idea behind it is rather strong, but unfortunately this committee tends to operate like a software 'club'—promoting the business ideas of its members and not general Linux principles, as it should. Having been created in 2009, nobody can see it in a big, serious project, though.

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